

Skill Week :- 2

Task 2:-

Code:-

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/types.h>

#include <sys/wait.h>


int main()
{
    char command[100];
    pid_t pid;


    printf("=====\n");
    printf("Enter Linux command (ls, pwd, date, whoami): ");
    scanf("%99s", command);


    pid = fork();


    if (pid < 0)
    {
        printf("Fork failed!\n");
        return 1;
    }


    else if (pid == 0)
    {
        // Child Process
        printf("\n----- Child Process ----- \n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n\n", getppid());
```

```

printf("Executing command: %s\n\n", command);

execlp(command, command, (char *)NULL);


// Executes only if exec() fails
perror("Execution failed");
exit(1);
}

else
{
    // Parent waits for child
    wait(NULL);

    printf("\n----- Parent Process ----- \n");
    printf("Parent PID : %d\n", getpid());
    printf("Child PID : %d\n\n", pid);

    printf("Child Process Finished Successfully.\n");
}

return 0;
}

```

Output:-

```
anuan@anushka:~$ gcc command_executor.c -o command_executor
anuan@anushka:~$ ./command_executor

=====
Enter Linux command (ls, pwd, date, whoami): pwd

----- Child Process -----
Child PID : 998
Parent PID : 997

Executing command: pwd

/home/anuan

----- Parent Process -----
Parent PID : 997
Child PID : 998

Child Process Finished Successfully.
anuan@anushka:~$
```